

blockMesh

-the basic OpenFOAm
mesh builder

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Note about Dimensions

- **OpenFOAM always works with a 3-D mesh**
- Even for 2-D and 1-D calculations, the mesh is generated in 3-D
- The directions which are not to be solved for are indicated by '**empty**' in the **normal** boundaries
- They should also have only **1** cell in this direction
- This indicates to the solver that the

Philosophy

- .Any domain can be broken into contiguous(no overlaps or holes) pieces called blocks
- .Blocks can be thought of as flexible cubes tracing the flow
- .When edges are defined as arcs, polyLines or splines, cube can become cylinder, sphere, etc
- .Each block has local coordinate system of x,y,z directions

Mesh Entities

- .Blocks are built step by step in blockMeshDict
- .**Vertices** are points/corners defined in space
- .**Edges** are defined by joining the vertices
- .**Blocks** are defined by combining eight edges
- .Type of cells (hex or tet), number of cells and their spacing(grading) is specified in block definition

Basic Examples

- The details used by '**blockMesh**' application/command are present in '**blockMeshDict**' file in '**system**'
- We will build a simple cube using blockMesh
- OpenFoam simple cavity tutorial file
- Command: `gedit`
`$FOAM_TUTORIALS/incompressible/icoFoam/cavity/cavity/system/blockMeshDict`
- We will then manipulate some entries to change the cube to cylinder
- Later we will see how several blocks can

blockMeshDict example

```

1 /*-----* C++ *-----*/
2 =====
3 \ \ / / F i e l d
4 \ \ / / O p e r a t i o n
5 \ \ / / A n d
6 \ \ / / M a n i p u l a t i o n
7 /*-----*/
8 FoamFile
9 {
10     version      2.0;
11     format        ascii;
12     class         dictionary;
13     object        blockMeshDict;
14 }
15 // *****
16
17 convertToMeters 0.1;
18
19 vertices
20 (
21     (0 0 0) //0
22     (1 0 0) //1
23     (1 1 0) //2
24     (0 1 0) //3
25     (0 0 0.1) //4
26     (1 0 0.1) //5
27     (1 1 0.1) //6
28     (0 1 0.1) //7
29 );
30
31 blocks
32 (
33     hex (0 1 2 3 4 5 6 7) (20 20 1) simpleGrading (1 1 1)
34 );
35
36 edges
37 (
38 );
39
40 boundary
41 (
42     movingWall
43     {
44         type wall;
45         faces
46         (
47             (3 7 6 2)
48         );
49     }
50     fixedWalls
51     {
52         type wall;

```

Scale

Vertices

Cell type Vertices No of cells in local x,y,z Cells spacing strategy Grading in local x,y,z

Block

Empty defaults to straight edges

blockMeshDict example

```
31 blocks
32 (
33     hex (0 1 2 3 4 5 6 7) (20 20 1) simpleGrading (1 1 1)
34 );
35
36 edges
37 (
38 );
39
40 boundary      All block faces should be here except internal faces
41 (
42     movingWall
43     {
44         type wall;
45         faces
46         (
47             (3 7 6 2)
48         );
49     }
50     fixedWalls      Boundary name
51     {
52         type wall;      Boundary type
53         faces
54         (
55             (0 4 7 3)
56             (2 6 5 1)      Block faces
57             (1 5 4 0)
58         );
59     }
60     frontAndBack
61     {
62         type empty;
63         faces
64         (
65             (0 3 2 1)
66             (4 5 6 7)
67         );
68     }
69 );
70
71 mergePatchPairs      Only needed if internal faces between two blocks are not same size
72 (
73 );
74
75 // ***** //
```


Run the case

.Run following commands in a terminal

```
.of7
```

```
.cp -r
```

```
$FOAM_TUTORIALS/incompressible/icoFoam/cavity/cavity  
$FOAM_RUN
```

```
.cd $FOAM_RUN/cavity
```

```
.blockMesh
```

```
.touch cavity.foam
```

.Open Paraview and drag cavity.foam into it

Local coordinate system

- .Blocks are defined by 8 vertices
- .For example `hex(0 1 2 3 4 5 6 7)`
- .0 1 2 3 forms one face
- .4 5 6 7 forms corresponding opposite face
- .Normal from 0 1 2 3 should point inward
- .Normal from 4 5 6 7 should point outward
- .Local x is along edge joining 0 1
- .Local y is along edge joining 1 2
- .Local z is along edge joining 0 4

Edit edges

- .We will now transform our cube to a cylinder
- .We will change edges from straight lines to arcs on faces (0 1 2 3) and (4 5 6 7)
- .Other options are 'polyLine' and 'spline'
- .Run the following commands
 - `.cp -r system/blockMeshDict system/blockMeshDict_cyl`
 - `.gedit system/blockMeshDict_cyl`

Edge as arcs

Edit the 4 5 6 7 vertices' z location from 0.1 to 1

Edit the edges sub-dictionary by adding lines as shown:

```
31 blocks
32 (
33     hex (0 1 2 3 4 5 6 7) (20 20 1) simpleGrading (1 1 1)
34 );
35
36 edges
37 (
38     arc 0 1 (0.5 -0.207106781 0)
39     arc 1 2 (1.207106781 0.5 0)
40     arc 2 3 (0.5 1.207106781 0)
41     arc 3 0 (-0.207106781 0.5 0)
42     arc 4 5 (0.5 -0.207106781 1)
43     arc 5 6 (1.207106781 0.5 1)
44     arc 6 7 (0.5 1.207106781 1)
45     arc 7 4 (-0.207106781 0.5 1)
46 );
47
48 boundary
49 (
50     movingWall
51     {
52         type wall;
53         faces
54         (
55             (3 7 6 2)
56         );
57     }
```

Run `blockMesh -dict system/blockMeshDict_cyl` in terminal

As practice can you now transform the cylinder to a sphere?

Cell spacing

- .We will make some changes to cell spacing to understand its effect
- .Cell spacing rule of thumb:have less spacing/more cells in regions of high gradients
- .'simpleGrading' is usually enough
- .User sets the ratio of first to last cell size
- .Intermediate cell sizes are calculated by GP

Change the cell grading

Edit the grading in blocks

```
31 blocks
32 (
33   hex (0 1 2 3 4 5 6 7) (20 20 1) simpleGrading (4 4 1)
34 );
```

But how to get small cell spacing in the middle?

Simple grading has an alternate format for specifying grading

It allows creation of sub blocks within each block

```
31 blocks
32 (
33   hex (0 1 2 3 4 5 6 7) (20 20 1) simpleGrading
34   (
35     ((0.5 0.5 0.25) (0.5 0.5 4))
36     ((0.5 0.5 0.25) (0.5 0.5 4))
37   1
38   )
39 );
```

Edge fraction Cell fraction Scale 2 brackets = 2 sub blocks along x and y

Grading in x Grading in x

Combining blocks

- .Domain can have several blocks
- .Some blocks will have common faces or face areas
- .The common areas must have same location of cells
- .If the cells do not match, we get 'inconsistent grading' error

Inconsistent grading example

- .Copy the file from
\$FOAM_TUTORIALS/compressible/sonicFoam/laminar/forwardStep to \$FOAM_RUN
- .Open blockMeshDict
- .Make changes in blocks as shown

```
39 blocks
40 (
41     hex (0 1 3 2 8 9 11 10) (25 10 1) simpleGrading (2 1 1)
42     hex (2 3 6 5 10 11 14 13) (25 40 1) simpleGrading (1 1 1)
43     hex (3 4 7 6 11 12 15 14) (100 40 1) simpleGrading (1 1 1)
44 );
```


You will obtain the following error:

```
Check topology

    Basic statistics
        Number of internal faces : 2
        Number of boundary faces : 14
        Number of defined boundary faces : 14
        Number of undefined boundary faces : 0
    Checking patch -> block consistency

Creating block offsets
Creating merge list

--> FOAM FATAL ERROR:
Inconsistent point locations between block pair 0 and 1
    probably due to inconsistent grading.

    From function void Foam::blockMesh::calcMergeInfo()
    in file blockMesh/blockMeshMerge.C at line 295.

FOAM exiting
```

Now modify blockMeshDict as follows:

```
39 blocks
40 (
41     hex (0 1 3 2 8 9 11 10) (25 10 1) simpleGrading (2 1 1)
42     hex (2 3 6 5 10 11 14 13) (25 40 1) simpleGrading (2 1 1)
43     hex (3 4 7 6 11 12 15 14) (100 40 1) simpleGrading (1 1 1)
44 );
45
```

As an exercise identify the common face between blocks 1 and 2 and change the grading

Hint: look at number of cells in each block along x, y and z

Review

- 'blockMesh' is the basic mesh builder in OpenFOAM
- It builds a mesh using contiguous blocks
- Each block can be specified using 8 vertices
- Vertex order is important and uses two opposite faces in same cyclic sense
- Grading can be used to obtain cell spacing in GP
- Edges can be set as arcs, splines or